

DATUM LINE

AI × BIO · 29 SEPT

Control is an illusion.

Confidence without control.

● your cells ○ the field

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Katherine Titterton

- 2023–2026 **Axiom Bio** SAN FRANCISCO
🏠 founding MTS • hepatocytes at scale
- 2022–2023 **Insitro** SOUTH SAN FRANCISCO
🤖 automation • assays for ML
- 2021–2022 **MIT** SYNTHETIC NEUROBIOLOGY
🔬 expansion microscopy
- 2018–2021 **AbbVie** NEUROSCIENCE
🧬 tau • neurons
- 2017 **Colgate University** BA
📖 neuroscience + math biology

MY WORK

- 1 Ewald JD, **Titterton KL**, et al. Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes. *Cell Systems*, 2026.
- 2 Seal S, Dee W, ... **Titterton K**, ... Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nature Communications*, 2026.
- 3 Wu JW, **Titterton K**, et al. A neuronal tau aggregation assay for tau-related drug discovery. *J Biol Chem*, 2025.

DATA AND METHODS

- 4 Chandrasekaran SN, et al. JUMP Cell Painting dataset. *bioRxiv*, 2023. All data here, CC0.
- 5 Bray MA, et al. Cell Painting. *Nature Protocols* 11:1757, 2016.

PUBLISHED WITH

Broad Institute • EMBL-EBI • HESI • Bayer • University of Cambridge • Uppsala University

“Do my cells look normal?”

IRREPRODUCIBLE PRECLINICAL RESEARCH · US · PER YEAR [6]

\$28B

36% ≈ \$10B · reagents & references

64% · design, analysis, protocols

1

search

do my hepatocytes look normal

images from a dozen papers,
no two protocols alike

2

ask around

To: whoever ran this line last
Subject: does this look right to you?
Attached: screenshot.png

the people who know are at other
institutions, and their images are not

3

run it anyway

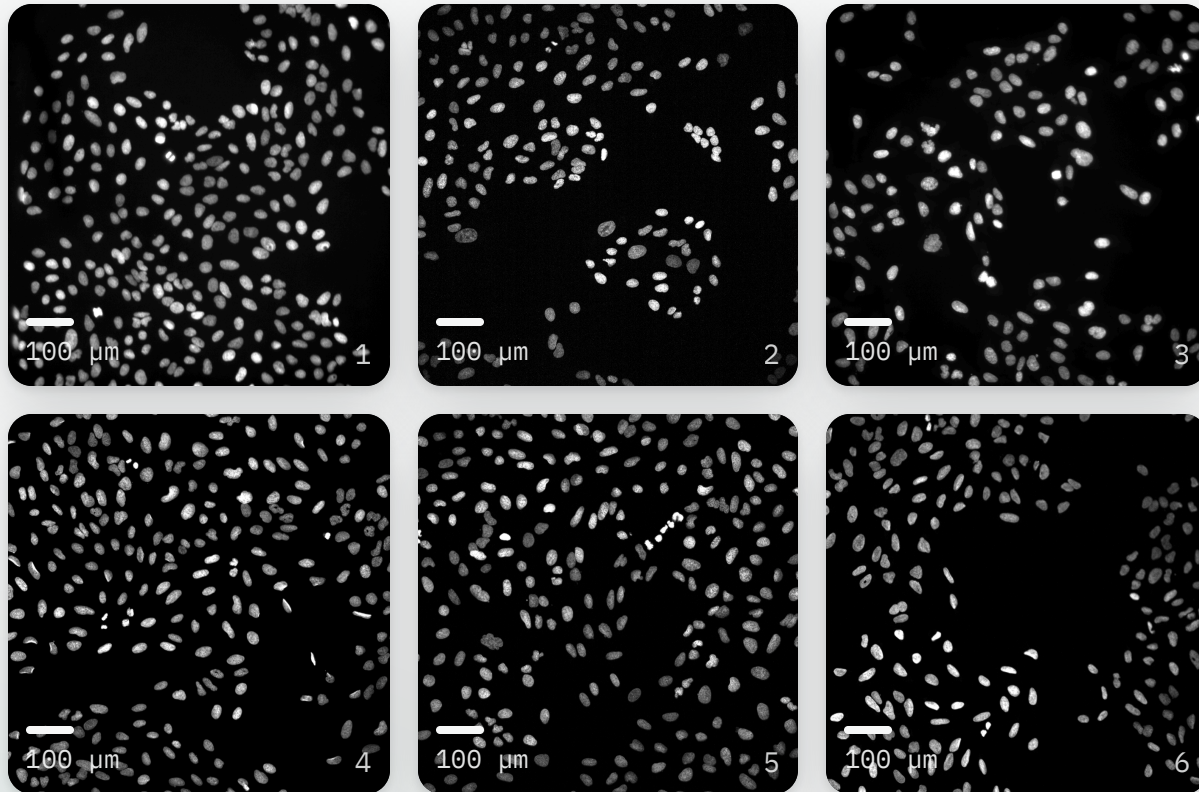
week 1 month 2 month 4

a screen at three runs a week,
about \$50k a run

\$2.6M of screening on cells
nobody had checked

Fig. 1 Checking a new line today: a workflow, not a measurement. 3 runs/wk × 17 wk × \$50k ≈ \$2.6M. [6] Freedman, Cockburn & Simcoe, *PLoS Biol* 13:e1002165, 2015.

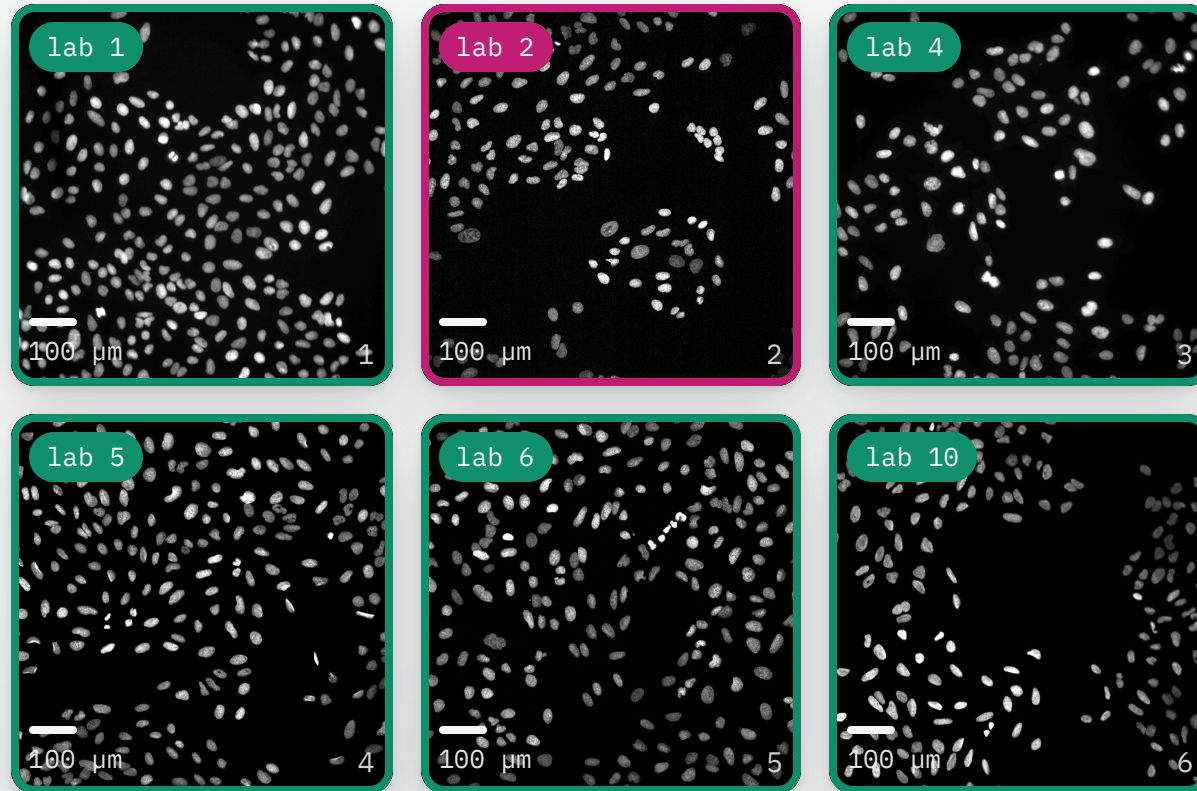
One lab, or six?



- 🧴 same line ✓
- 📋 same protocol ✓
- 🚫 no drug ✓

Fig. 3 JUMP cpg0016 (CC0), untreated wells, DNA channel · n = 12,046, 7 labs · 96.2% vs chance 14.3%, held out by plate

Could you tell? **A classifier can.**



96% CLASSIFIER

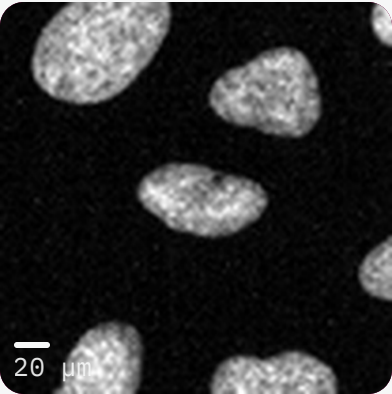
14% CHANCE

□ the 4% it gets wrong

Fig. 3 JUMP cpg0016 (CC0), untreated wells, DNA channel · n = 12,046, 7 labs · 96.2% vs chance 14.3%, held out by plate

One well, all the way out.

ONE NUCLEUS



192 μm² · 0.59

ITS WELL

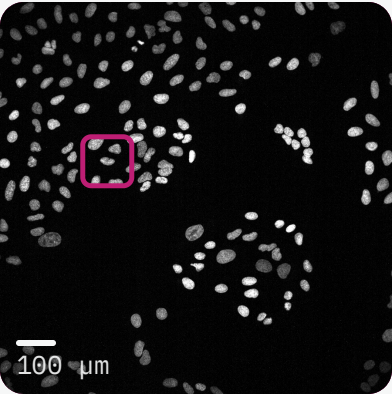
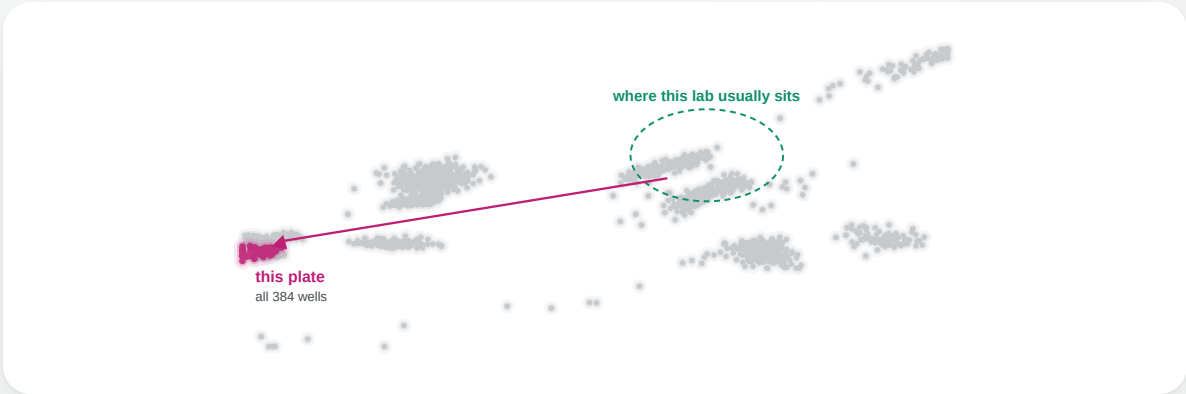


plate 1053600681

3,781
numbers

→
one point

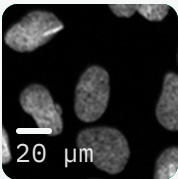
WHERE IT LANDS



12,046 wells, 7 labs

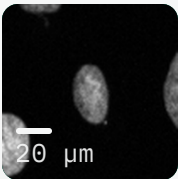
other labs

lab 5



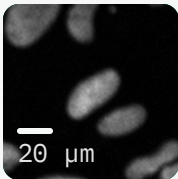
195 μm² ·
0.94

lab 6



213 μm² ·
0.50

lab 10



207 μm² ·
0.29

AREA ≈ ≈
TEXTURE ≠ ≠

One lab's normal is not another's. The difference is shape, not count.

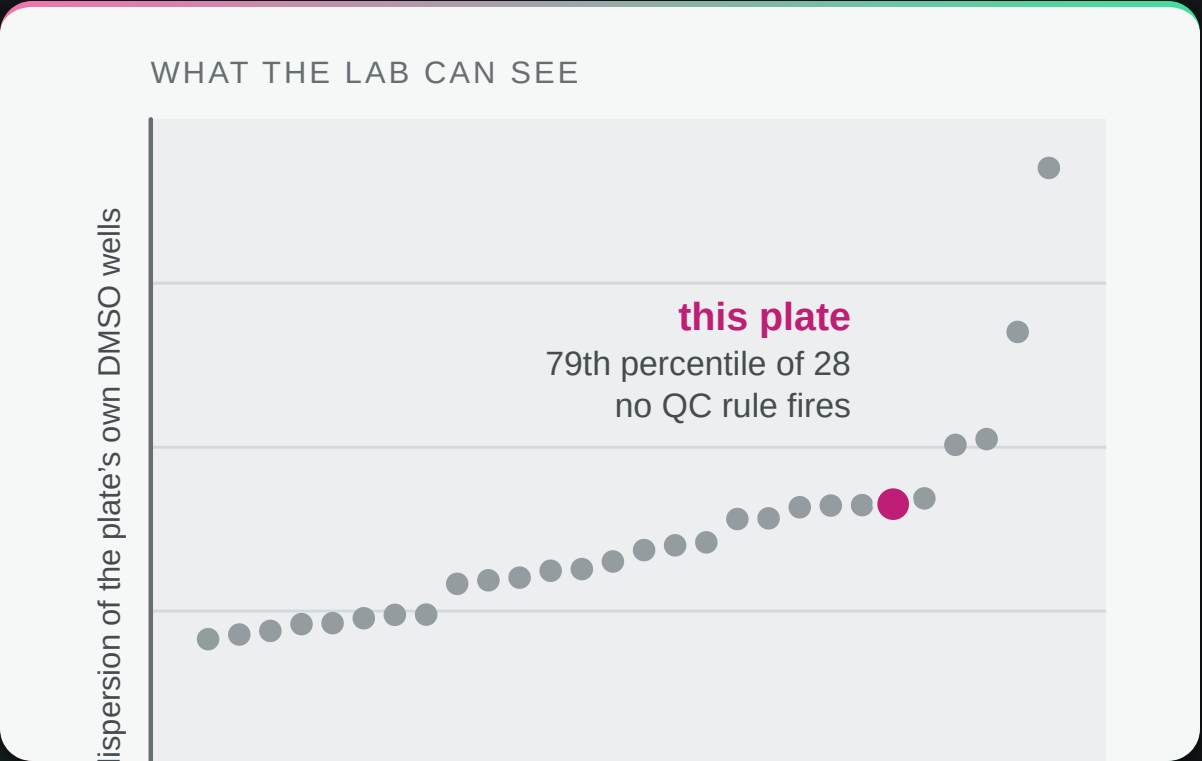
Fig. 3 JUMP cpg0016 (CC0), untreated wells, DNA channel · n = 12,046, 7 labs · 96.2% vs chance 14.3%, held out by plate

It passed every check.

plate 1053600681

WHAT ITS OWN LAB CAN SEE

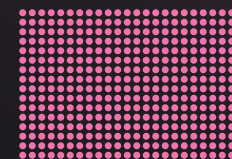
✔ local QC passed



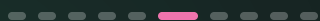
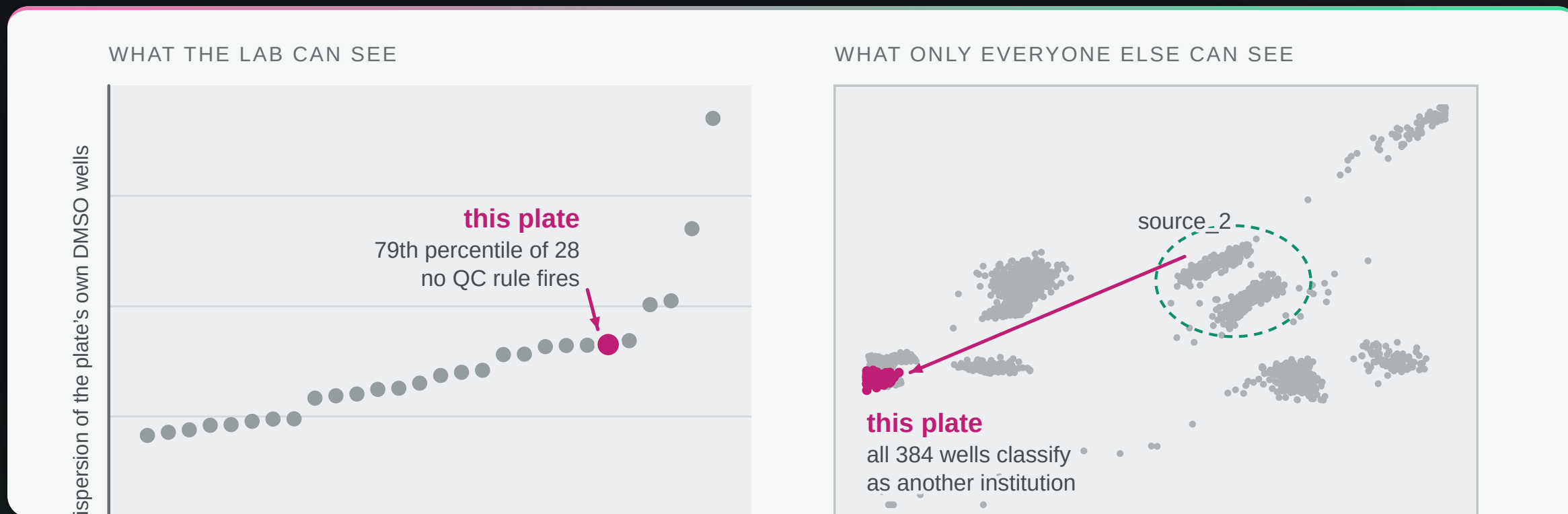
Within-plate spread: 79th percentile of 28. Click for what only the field can see.

It says lab 2.
Its cells say otherwise.

96%

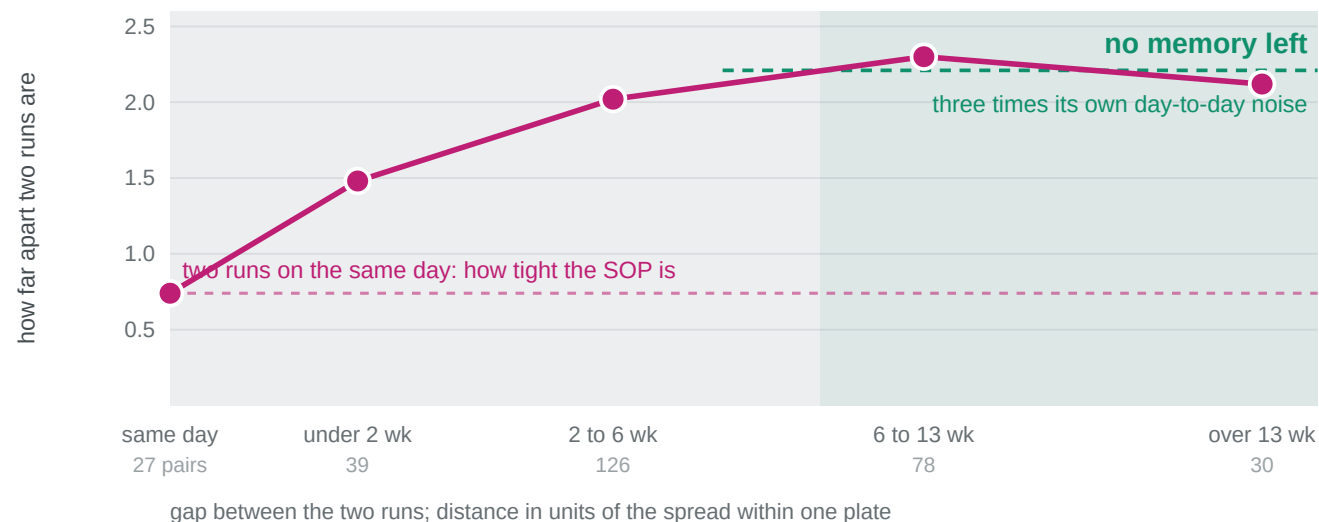


```
384 wells · 0 say lab 2
```



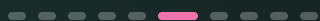
Same cells, same SOP, same building.

3x

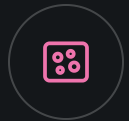


One institution · 31 plates · 8 runs

Your own past is not a reference. The field is.



Know on day one, **not month four.**



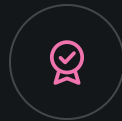
brightfield
control wells



stays
yours



placed vs
the field



answer
in a day

WHO IT IS FOR • WHAT IS AT STAKE



ACADEMIC LABS AND CORES
the paper



PHARMA AND CRO SCREENING
the screen • ~\$2.6M



CELL BANKS AND VENDORS
the lot

today



4 months

datum line ● 1 day



Certificate

FOR THE RUNS SUBMITTED



YOUR CELLS

primary human hepatocytes, lot and passage

COMPARED WITH

every lab running this line, and the published set

RESULT

in distribution

outside it

Held by nobody who sells you cells, media or the instrument.

Identity, yes. Behaviour, no.

	STR AUTHENTICATION	VENDOR CERTIFICATE	THE PLATE'S OWN CONTROLS	PHENOMICS PLATFORMS	DATUM LINE
Which line it is	✓	✓	—	—	run STR
How it behaves, the whole cell	—	—	✓	✓	✓
Against every lab running the same line	—	—	—	removed on purpose	✓
From images you already take	—	—	✓	—	✓

Identity is solved.

Behaviour is not.

Every institution onboarding a line has this problem: academic cores, pharma and CRO screening groups, cell banks and reagent vendors.

	STR	PLATE QC	US
proves which cell line you have	✓	.	✓
checks the cells against your own plate	.	✓	✓
checks them against every other lab	.	.	✓
says which way to move them back	.	.	✓
flags a bad reagent lot across labs	.	.	✓

STR is a collaborator, not a rival. Nobody sells the behaviour half.

Plate QC: within-plate control checks offered by instrument vendors. Phenomics platforms remove lab-to-lab differences by design, which is the opposite operation.

Build the ugliest dataset in biology, **on purpose.**

Every method here needs data that contains the failures, not data that avoids them. Nobody has built that on purpose.

THE TEST SET

Variation, deliberately

Multiple lines and plate types, single planes and stacks, different objectives, damaged plates: the variation a real lab meets, generated rather than avoided.

RED TEAM

Break it on purpose

Corrupt the metadata, defocus the objective, clip the well edge. The model has to say not just that something is wrong but which thing.

THE PAYOFF

A taxonomy of failure

Air bubble, focus failure, clipped well, bad lot. Outliers named, not just flagged, so a lab knows what to fix.

A foundation other people can stand on: one phenotypic space the field converges toward, instead of Google Images.

Every plate you image, placed in the field.

WHERE THIS GOES



NEXT

what to change, which lots work



THE SET

a reference built to break



THEN

bad-lot alerts across labs



BEYOND

microscopes · catalogue · drift

WHAT I NEED



Images U2OS controls



People ML, normalization



Bench 3-site proof, then the set



Pre-seed to run it

Have images of U2OS cells? Talk to me.

The rest, if you ask.

Find the question, jump to the board. Every number traces to results/e1, e1b, e2 or a numbered reference; pending work says pending.

A1 **Why not trust the plate’s controls?**
There is no control well

A2 **Show me the math.**
Six operations on one embedding

A3 **Why not batch-correct it away?**
What JUMP records

A4 **Is the model the moat?**
Counting cells often wins

A5 **Why not just count cells?**
Two ways to ask one question

A6 **How often do I re-certify?**
The line is not expired

A7 **Isn’t this STR?**
STR says who, images say how

A8 **Is the assay itself working?**
A known answer

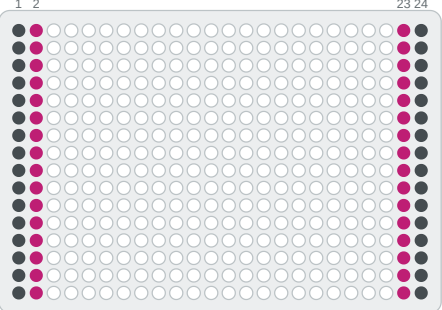
A9 **Does it change the result?**
Same drug, different answer

A10 **Does it work without stains?**
Bet on brightfield

There is no control well.

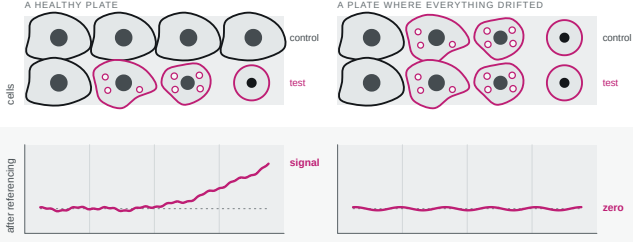
Every result is test minus control, on the same plate. But a whole plate can be off, or a whole batch, or a whole institution. A control only sees inside its own box.

A REAL JUMP PLATE · 384 WELLS



● DMSO 32 ● positive 32
○ compounds 320

The pink wells are the plate's zero: 32 of 384, near the edge by design.



A control only sees inside its own box.

Whole plate off: its controls move with it.
Whole batch off: so do every plate's.
Whole site off: nobody in the building can tell.
Only the field sits outside every box.

$$x_{test} = \mu + S + \tau(S) \quad x_{ctrl} = \mu + S \quad \Delta = \tau(S)$$

S , the plate's state, cancels from the number. Never from the cells.

So a control only sees inside its own box. If the plate is sick, so is the control. Then how would anyone know?

Fig. 2 Where a control sits, and what it can see. Left, a real 384-well JUMP compound plate (source_2, 1053601756): 32 DMSO, 32 positive, 320 compound wells. Right, schematic cells and traces, and the nesting of well, plate, batch, site and field. No measurements shown.

Six operations on one embedding.

Every well becomes a vector $e \in \mathbb{R}^{3781}$, z-scored across the field. Everything on the previous slides is one of these, and each one is a difference of means or a held-out classifier, so a lab can check it.

1 · EMBEDDING

[1]

$$e = \varphi(\text{image}); \varphi \in \{\text{CellProfiler}, \text{CNN}, \text{ViT}\}$$

Which φ barely matters: the three performed similarly on hepatocytes. The extractor is not the asset.

2 · LAB CLASSIFIER

96%

$$f : e \rightarrow \text{lab}, \text{ held out by plate}$$

Model and folds: [results/e1]. Balanced accuracy, chance 1/7. Permuted labels 15.2%. Without the 1536-well source: still 96%.

3 · DISTANCE, THE CERTIFICATE

[e6]

$$d = \| E[e \mid \text{plate}] - E[e \mid \text{field}, c] \|_{\sigma}$$

Difference of two means in units of the field's spread, conditioned on c . Its direction is what moved; $-d$ is the way back.

4 · VARIANCE DECOMPOSITION

[e5]

$$\text{Var}(e) = V_{\text{model}} + V_{\text{site}} + V_{\text{batch}} + V_{\text{plate}} + V_{\text{well}}$$

Nested, so shares sum to one. Microscope model separates from site because models are shared across sites.

5 · REGRESS VARIABLES OUT

12%

$$e_{\perp} = e - \hat{E}[e \mid c], \text{ then re-run } f$$

Remove only what you know is instrument, keep what may be biology. Remove the whole batch, as reference alignment or covariance matching does, and f collapses.

6 · MORPHOLOGY CHANGE, TWO WAYS

 δ

$$\begin{aligned} \text{DISTANCE } \delta &= f(\|e - E[e \mid \text{vehicle}]\|), f \text{ bounded} \\ \text{CLASSIFIER } g: e' &\rightarrow \text{label}, e' \text{ aligned to the reference before comparison} \end{aligned}$$

Same vector, two questions. Discovery wants invariance to the lab: align first. QC wants the lab coordinate: never align it away.

results/e1, e1b, e5, e6 · [1] Ewald et al. 2026

What JUMP records. What nobody does.

Normal is a conditional expectation, $E[x \mid \text{every variable below}]$. Even JUMP, built to be comparable, records only the instrument side.

RECORDED IN JUMP

site batch date plate
plate type well position
microscope confocal or widefield
laser or LED objective NA
sites per well filter set
pixel size brightfield planes
CellProfiler version

NOT IN ITS PUBLIC METADATA

DMSO % per well DMSO lot
DMSO water uptake stock freeze-thaw
serum lot media lot passage thaw date
seeding density confluence at dosing
dosing to fixation stain and dye lots
operator incubator shelf
edge evaporation mycoplasma status
CO2 and humidity

NORMALISE THE LAB AWAY AND THE LAB
DISAPPEARS

96%

raw features

balanced accuracy naming
the lab

14%

per-plate standardised

at chance, 1 in 7

12%

per-batch whitening
(TVN)

below chance: the
coordinate is gone

Same 12,046 wells, same held-out-by-plate
split. results/e1.

Per-batch whitening, the standard fix, names the lab 12% of the time: below chance. It deletes the coordinate a certificate needs.

Batch correction is built to remove this signal: Arevalo J, et al., Nature Communications, 2024. Variables: JUMP datasets metadata.

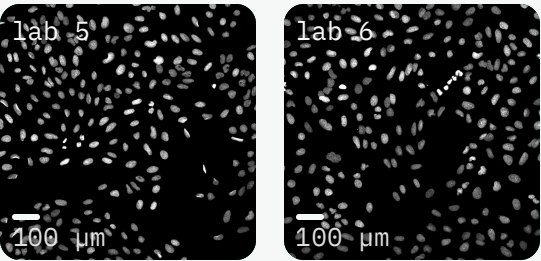
And the benchmarks are so weak that **counting cells** often wins.

If a naive baseline matches your model, the benchmark is the problem. The bottleneck was never the model. It is what you are comparing against.

Two ways to ask the same question.

Why now: an image classifier can answer this, and until recently nothing could. A count moves only when cells die or divide; an embedding of the whole well moves when anything changes.

SAME CELL LINE, SAME PROTOCOL



cell count	236	224	same
shape	0.94	0.50	different

METHOD ONE

Does this run belong with the past runs?

$g: e' \rightarrow \text{belongs / does not}, \quad e' = \text{align}(e)$

Align first, so the answer is the cells and not the day.

Standard alignment; the tuning is ours.

METHOD TWO

How far out, and which way back?

$\delta = f(\|e - E[e \mid \text{vehicle}]\|), \quad f \text{ bounded}$

Bounded, so one wild well cannot swamp the score.

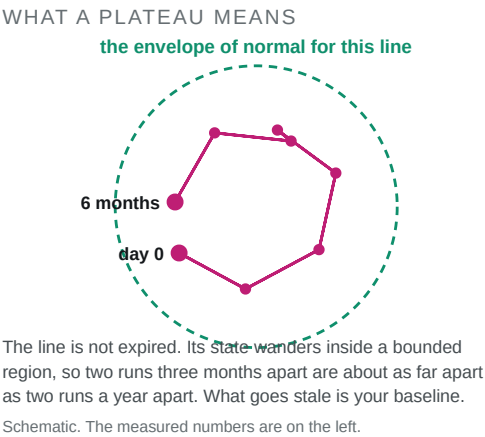
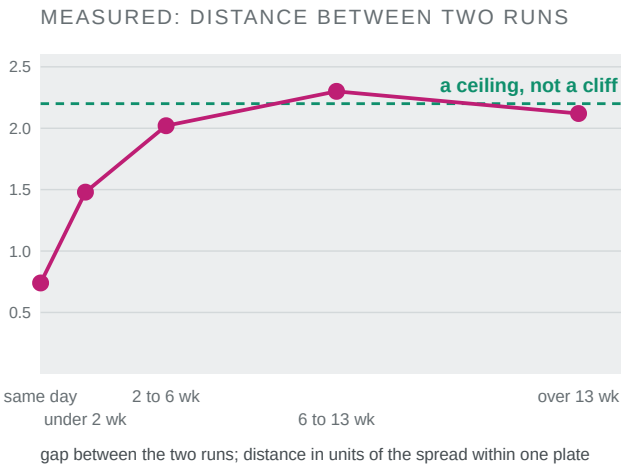
Distance published with the Broad [1]; transform unpublished.

Both methods need the same thing to work: somebody else's images. It takes a reference to know which.

Fig. 1 Same cell line, two untreated wells, and the two readouts. Left, DNA channel, JUMP cpg0016, CC0; nuclei by Otsu segmentation; detail = high-frequency energy over total, inside each nucleus. Right, the two readouts: a classifier on aligned embeddings, and a bounded distance from the vehicle centre. Alignment and distance follow published practice [1]; the specific transform and tuning are not disclosed here. [1] Ewald JD, Titterton KL, et al. *Cell Systems*, 2026.

The line is not expired. Your baseline is.

Two runs of the same cells drift apart for about three months, then stop. Drift is bounded, not runaway, which is exactly why a reference is possible at all.



IS THE CELL LINE RUINED?

No. A plateau means the state stays inside an envelope. Cells three months old are not on their way somewhere worse.

SO WHY RE-CERTIFY?

Because your certificate is a comparison made on one day. By three months it is as stale as it will ever get, so a quarter buys the whole useful life of one reading.

HOW SOLID IS THE NUMBER?

One institution, 31 plates, 8 runs, June to October. The last bin is 30 pairs, so 2.30 then 2.12 is a plateau, not a decline. Every partner re-measures it.

Distance between plate centroids, scaled by the spread of wells within a plate; matched well counts. Same-day pairs sit at 0.74 (n=27); runs months apart at about 3.1x that. results/e1b.

STR says who. Images say how.

Not a competitor: the other half of the same certificate. Every lab already runs the left column. The right column costs nothing extra.

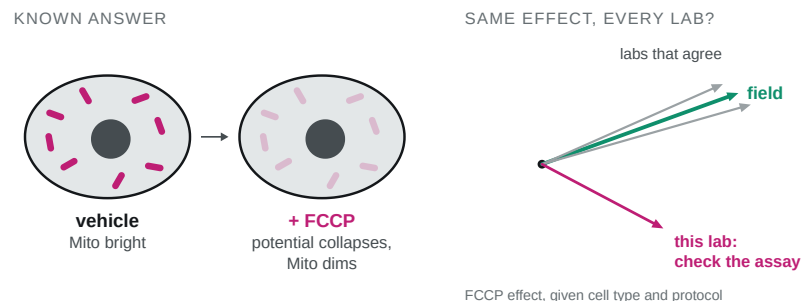
	STR PROFILING	CONTROL-WELL IMAGES
WHAT IT READS	DNA at 17 short tandem repeat loci	the pictures the screen already took
WHAT IT PROVES	the line is what the label says	the cells still behave like the field's version of that line
WHAT IT MISSES	two cultures, same fingerprint, different behaviour	identity: it cannot tell you the line is mislabelled
SAMPLE	cells spotted on a card and mailed	nothing leaves the building but images
EXTRA WORK	a separate prep, per line, per test	none: control wells you already image
CADENCE	on receipt, after 10 passages, when in doubt	every quarter, from the last plate you ran
ANSWER	a profile to compare by eye or by database	one of two words, and a distance

Identity once. Behaviour every plate. One line on the certificate each.

STR practice and cadence: ATCC cell authentication service and test recommendations; 17 loci plus amelogenin on the Promega PowerPlex 18D system. Low- and high-passage LNCaP respond differently to the same hormones: ATCC technical document.

Next: a known answer, run in every lab.

Controls certify the cells. Positive controls certify the assay. Once cell type and protocol are accounted for, a known perturbation should move cells the same way everywhere.



RUNS NOW ON JUMP

Compound plates already carry 8 positive controls in 32 wells. Same effect at every site?

ADD ONE: FCCP

MitoTracker loads by membrane potential; FCCP collapses it. Intensity is the obvious readout and the worst one: gain, exposure and filter set differ by lab. The shape change does not.

THE TEST

$$\delta = E[x \mid pos, c] - E[x \mid vehicle, c]$$

Pass if a lab's δ points where the field's does, at a size inside the field's spread. Not “did the channel dim” but “did the cells change the way the field says they change”. Vehicle fails: fix the cells. Positive control fails: fix the assay.

FCCP or CCCP as the loss-of-potential control: Thermo Fisher MitoTracker guidance. MitoTracker Deep Red dims on depolarisation: PMC4811937. Intensity is not comparable across instruments without calibration; morphology is scale-free. Response differs by cell type: Pendergrass et al., Cytometry A, 2004. Plate layout: JUMP metadata.

Same drug. Different answer?

Every JUMP compound plate runs the same eight positive controls beside its DMSO. So the field already contains the experiment: one fixed compound, measured 1,829 times, in eleven buildings.

WHAT A SCREEN REPORTS, PER PLATE

$$\delta_{p,c} = E[x \mid \text{plate } p, \text{ compound } c] - E[x \mid \text{plate } p, \text{ DMSO}]$$

Treated minus control, on the same plate. Scaling fit on the field's DMSO wells only, never on the plate being judged.

Q1 · DOES IT AGREE?

[e8]

Cosine between each plate's δ and the field's, within a lab against across labs.

AMG900, dexamethasone, FK-866, LY2109761, NVS-PAK1-1, quinidine, TC-S-7004, aloxistatin.

Q2 · DOES IT TRACK THE CONTROLS?

[e8]

Spearman between a plate's control distance and how far its δ sits from the field's.

The claim: the further your controls sit, the more your answer moves. Null: shuffle plates within lab.

Q3 · DOES IT MATTER?

[e8]

Across-lab spread of $\|\delta\|$ as a percentage of the effect itself.

The sentence for the slide: the same compound, measured X% differently depending on the building.

If Q2 is flat, the pitch is that your cells differ. If Q2 is positive, the pitch is that your answers differ.

1,829 compound plates carrying both poscon and DMSO, 11 sources, 129 batches; every compound on at least 1,730. 144,279 wells. Positive controls: JUMP perturbation_control.csv. Scripts: analysis/e8_pull_wells.py, analysis/e8_same_drug_different_answer.py.

Shown in fluorescence. Bet on brightfield.

Every number here is Cell Painting: five stains. The product reads brightfield: none. That gap is the bet, testable on public data.

WHAT IS SHOWN

Stained wells name the lab

Vehicle-only U2OS wells, seven institutions: the lab is named 96% of the time, chance 14%. results/e1

WHY THE BET IS FAIR

Fewer variables, not more

No stain lot, dye lot or staining step: three unrecorded variables gone (A3). JUMP images brightfield beside the stains, so the same wells re-run without them.

WHAT WOULD KILL IT

Brightfield alone at chance

Then the certificate asks for one cheap stain. The test: e1 on the brightfield planes of the same 12,046 wells.
[pending]

Brightfield timecourse is the modality I build screens on.